

Quantum Mechanics III

Lecture 13

Quantum Process Tomography

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$p_{jk} = \text{Tr}[(\rho_j^T \otimes E_k)J_{\mathcal{E}}]$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$J_{\mathcal{E}} \geq 0, Tr_{out} J_{\mathcal{E}} = I$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$QDC_{new} = QDC_{26} + \Delta J$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 13; QPT Laboratory Guide; QDC diff standard